2026-04-14

Clustering Techniques

K-Means & Hierarchical Clustering

iXPERIENCE

GO FURTHER

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Contents

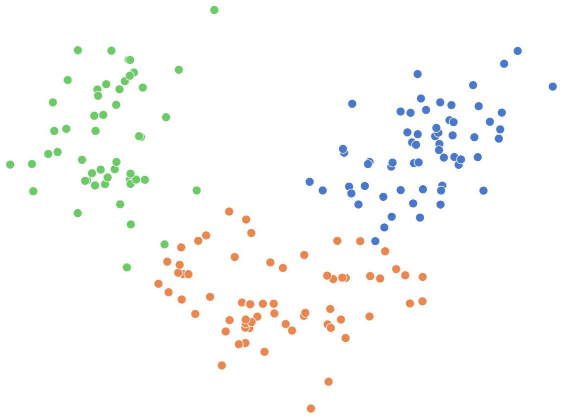
What is Clustering?

- Unsupervised learning: no labels, no known right answer. The algorithm finds structure on its own
- The goal is to discover natural groupings: customer segments, document topics, anomaly types
- Unlike classification, you don't know the groups in advance: you're exploring the data

K-Means Clustering

The Idea

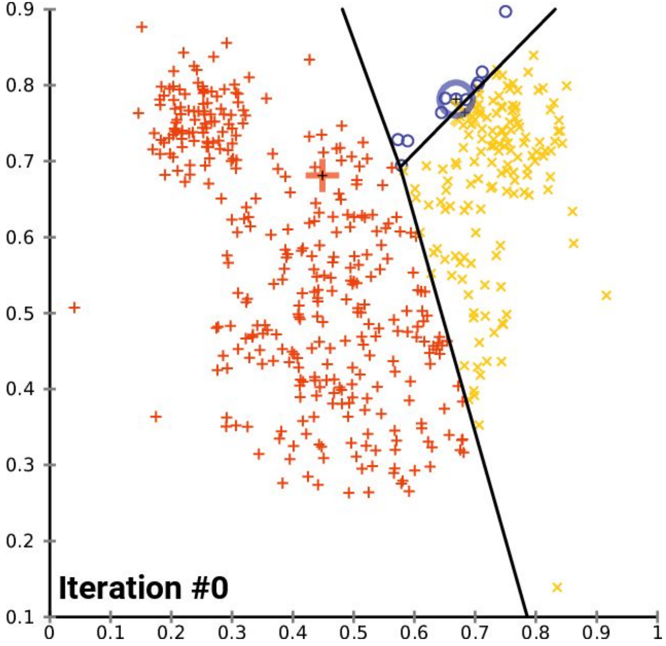
Group data into k clusters by minimizing the distance from each point to its cluster center (centroid)



K-Means Clustering

Steps

1. Pick k random points as initial centroids
2. Assign every point to its nearest centroid
3. Recompute each centroid as the mean of its assigned points
4. Repeat steps 2–3 until assignments stop changing



K-Means Clustering

What you need to know

- You must choose k before fitting
- Sensitive to feature scale (always use `StandardScaler` first)
- Starts from a random initial state (runs multiple times to avoid bad starts)
- Objective: minimise **inertia** (total within-cluster variance)
- Time complexity: ~O(n)

Choosing k

Elbow method + Silhouette score

The Elbow Method:

- Plot inertia vs. k and look for the "elbow".
- Inertia is the total sum of squared distances from each point to its centroid — lower means tighter clusters.
- The elbow is not always obvious, so always use a second method to confirm.
- Limitation: the elbow can be subtle or ambiguous

The Silhouette Score:

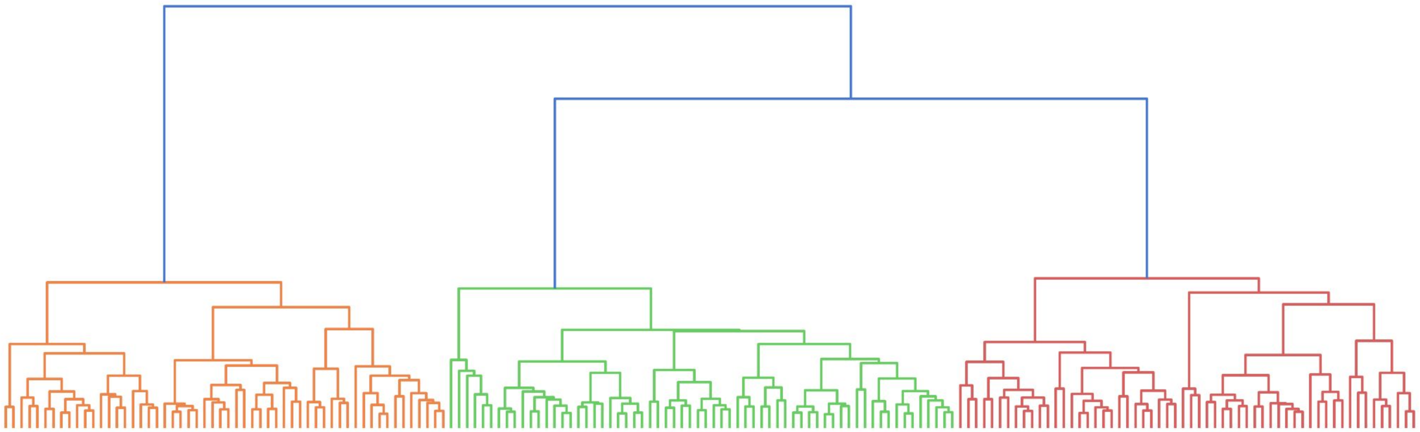
- Measures how well each point fits its own cluster compared to the nearest other cluster.
- Ranges from -1 to 1 (higher is better).
- Pick the k with the highest silhouette score.

Use both methods together.
If they agree → confident choice. If they disagree → look at the data.

Hierarchical Clustering

The idea

Build a tree of merges, no need to specify k upfront. Start with every point as its own cluster, then merge the two closest clusters repeatedly until one remains.



Hierarchical Clustering

Linkage: how do we measure "closest"?

Linkage	Distance between clusters	Distance formula
Ward	Minimizes increase in total variance (best in practice)	$d(A,B) = \frac{\lvert A\rvert\,\lvert B\rvert}{\lvert A\rvert + \lvert B\rvert} \lVert \mu_A - \mu_B \rVert^2$
Complete	Minimize maximum distance between any two points	$d(A,B) = \max_{x\in A,\, y\in B} d(x,y)$
Single	Minimize minimum distance between any pair of points (good for “stringy” clusters)	$d(A,B) = \min_{x\in A,\, y\in B} d(x,y)$

The Dendrogram

What is it?

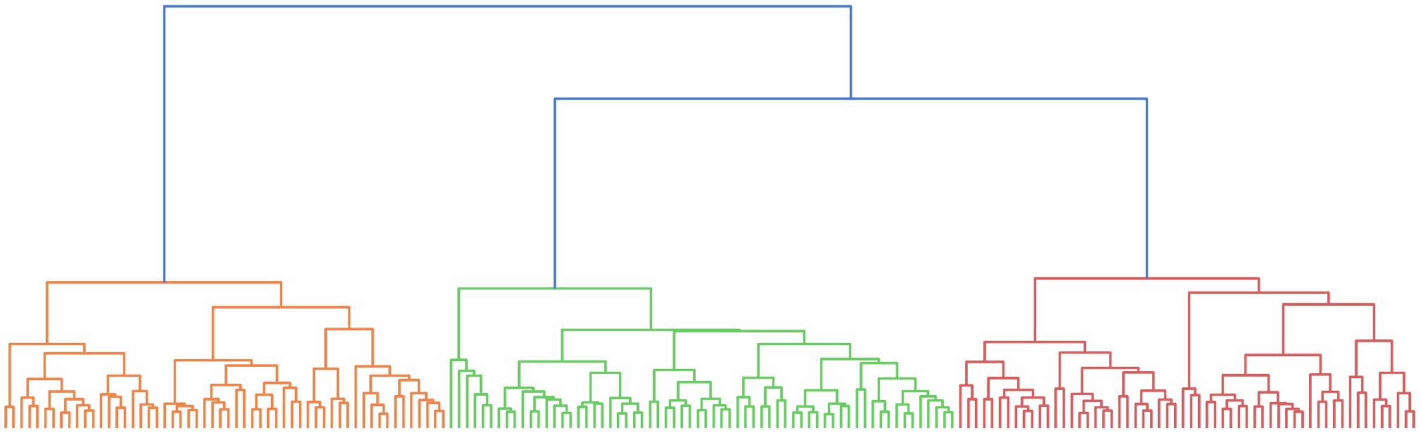
A tree diagram showing the full merge history.

- Bottom: individual points
- Height of each branch: how dissimilar the two merged groups were
- Top: everything merged into one cluster

The Dendrogram

How to read it

Draw a horizontal line across the diagram. Count the vertical lines it crosses: that's your k.



K-means vs Hierarchical Clustering

	K-means	Hierarchical Clustering
Specify k upfront?	Yes	No
Best for	Large datasets, spherical clusters	Smaller datasets, unknown structure
Time complexity	$O(n) \rightarrow$ fast	At least $O(n^2) \rightarrow$ slow on large data
Reproducible?	No (random init, use n_init)	Yes
Sensitive to outliers?	Yes	Less so
Visualization	Centroid-based scatter	Dendrogram

Go_Let’s code!

Key takeaways

- Clustering is **unsupervised**: no answers to look up
- **K-Means**: fast and interpretable
- **Hierarchical**: more exploratory
- Always **scale features** before clustering, distance methods are unit-sensitive
- Both methods recovered three chemically distinct wine cultivars with ~90% accuracy from raw measurements, with no labels
- In practice: run both, compare metrics, and ask which result makes the most **domain sense**